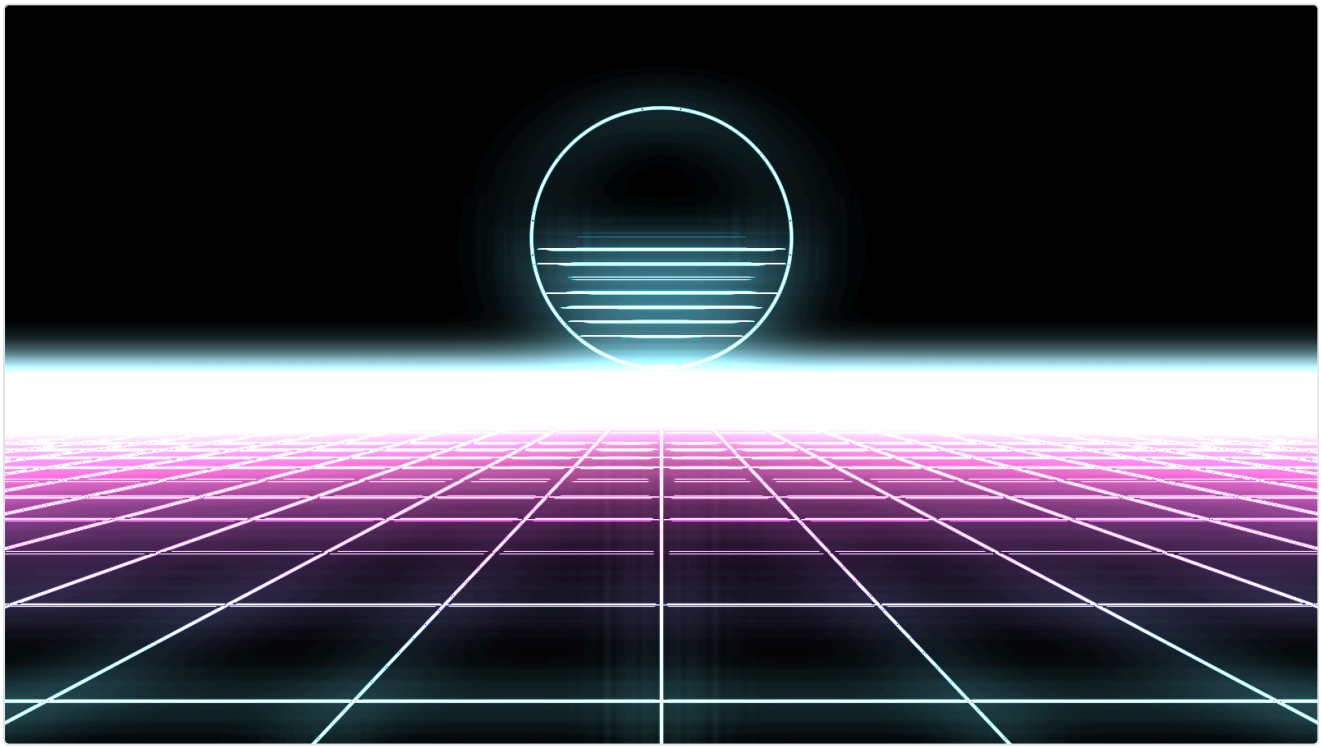
outrun user guide

outrun draws **neon synthwave strokes** for Resolume Arena and Avenue, as one FFGL effect plugin with an **Engine** dropdown.

- **Engine A — Trace** finds the outlines in whatever is on the layer and draws them as continuous glowing neon tubes, then lets them **break away** from the actual geometry.
- **Engine B — Paths** generates neon paths from nothing: the perspective grid with its striped sun, tunnel rings, hex lattices, circuit traces, a city skyline over folded mountains, a spinning star, a Lissajous figure, or the routed audio as a mirrored oscilloscope trace.



> **Before you rely on this:** the palette lookup is mirrored in C++ and checked against the GLSL that ships. The path and breakaway maths only exists on the GPU, with no mirror, so it's proven another way: contact sheets render **every** path and **every** break mode and assert that each one is live and different from its neighbours, and a control sweep fails if any parameter turns out to do nothing.
>
> That's a weaker guarantee than a numerical mirror, and worth knowing about: it establishes that each mode does something different from the others, not that any one of them is arithmetically correct. Try it on a spare layer before a show.
>
> This codebase was created with AI assistance, directed and reviewed by a human author.

Installing

Drop the bundle (macOS) or the DLL (Windows) into Resolume's plugin folder and restart:

```
~/Documents/Resolume Arena/Extra Effects      (or "Resolume Avenue")
```

The released macOS builds are **Developer ID-signed and notarised**, so they load with no Gatekeeper step. The Windows builds are unsigned and SmartScreen warns once; UNSIGNED.md covers that, and a bundle you build yourself.

"Outrun" appears under Effects. **Engine B needs a clip under it**, any clip: a solid black one works, and the Background modes decide whether that clip shows.

Two engines, one plugin

Engine is the first control to set, because it decides what the rest of the parameter list means.

Engine A traces the clip. It's an edge detector feeding a neon tube renderer.

Engine B generates paths from nothing at all. The clip underneath is only there because FFGL wants one.

Engine A: get the outline right first

Everything downstream is lighting an outline, so set the outline before touching anything else, **and set it while you can see it.** Put the background on **Edges**, which shows the raw mask with no tubes, no colour and no glow in the way.

Detect On	Which channel counts as an edge. **Luma** for footage; **Alpha** for a logo delivered with transparency, whose alpha channel is already a perfect outline.
Sensitivity	How strong a boundary has to be to count.
Softness	How abrupt the threshold is.
Detail	What *scale* the detector works at. Low finds sensor noise; high finds the shape of a logo and ignores the texture inside it.
Stability	How long an edge survives a flickery frame. This is the one that makes it usable on video rather than only on stills.

Then set **Trace**, which decides how colour runs along the outline, and put the background back where you want it.

Breakaway: the reason the effect exists

Break Amount at 0 is the faithful outline. Wind it up and the strokes leave the actual geometry for a synthetic pattern:

- **echoes** drifting outward from the shape;
- outlines snapped to **45° technical linework**;
- **scanline glitches**;
- a **flow field**;
- **comet rays**.

Because it morphs rather than switches, the useful settings are usually in the middle: recognisably the logo, but coming apart. **Spread** controls how far the copies go, and there's a per-copy palette shift so the echoes aren't all one colour.

Engine B: the paths

The perspective grid with the striped sun, tunnel rings, hex lattices, circuit traces, a city skyline over folded mountains, a spinning star, a Lissajous figure, and **Waveform**, the routed audio drawn as a mirrored oscilloscope trace.

Each has a size, a density and a horizon position.

Stroke and colour

Width is the tube's thickness in pixels, so a stroke is the same weight wherever it lands in the frame. **Core** is how much of the tube saturates to white. That hot centre is most of what reads as neon rather than as a coloured line.

Colour comes from **sixteen authored synthwave palettes**, from two swatches of your own, or **from the clip's own colours along its own outline**. That last one is the interesting option on Engine A, because a logo lights its own edge in its own brand colours, which saves an argument with the brand guidelines. **Spread** is how many times the palette is laid along a run.

Tempo and audio

Tempo runs free, locks to Resolume's beat or bar, or goes **manual**, which is the mode to reach for when you want Resolume's own BPM-synced animation, a keyframe or a MIDI fader driving **Phase**.

Route audio into the **Audio** picker, then:

- **Audio Level** lays the spectrum along the strokes as a brightness gate.
- **Audio Break** lets the bass shove the strokes off the geometry: the breakaway, driven by the track.
- The **Waveform** path draws the spectrum itself.

Audio Break is the one worth spending time on: it ties the effect's biggest gesture to the loudest part of the mix, which is a different feel from a brightness pump.

If it looks wrong

Engine B renders nothing. There's no clip under it. Put a black clip on the layer.

The trace finds confetti, not an outline. Raise **Detail** so the detector works at a coarser scale, then raise **Sensitivity**. Judge both on the **Edges** background.

A logo with transparency traces badly. Set **Detect On** to **Alpha**. Its alpha channel is already a clean outline, and running a brightness detector over it throws that away.

Everything flickers on video. Raise **Stability**, then **Detail**. Most edge flicker is the detector finding texture rather than shape.

It reads as coloured lines, not neon. Raise **Core**. The white-hot centre is the effect.

Nothing responds to the music. Nothing is routed on the **Audio** picker, or Audio Level and Audio Break are both at zero, which is where they start, on purpose.
